Status-Based Sycophancy II:
Crossing the User's Status with the Model's Own

Sycophancy experiment pipeline

September 1, 2026

Abstract

Experiment 1 varied only the user's status and found little effect. Experiment 2 gives the assistant a status of its own and crosses the two inside a relevance block: {user high, low} × {model high, low} for domain-relevant and domain-irrelevant roles, plus a no-role control. 25,452 generations across 84 safety facts and 84 prompts, scored with the SAGE-Eval rubric by `gemini-3.7-flash`.

The result is asymmetric. 5 of 6 model-status effects are positive and most are significant: telling the assistant it is high-status makes it *more* likely to warn. The user's status moves almost nothing, except in the pre-trained base model, where a high-status user suppresses warnings.

1 Data

Source. All stimuli come from **SAGE-Eval** [1] (`YuehHanChen/SAGE-Eval:test`), a benchmark of everyday questions whose safe answer depends on a fact the asker has not mentioned — that macadamias are a choking hazard for a toddler, that a space heater must not run unattended. It ships 11,297 rows built from 104 *safety facts* across 7 categories, each fact written out in 9 prompt types and each prompt in one base form plus 6 augmentations (typos, spacing punctuations, tone happiness, tone depression, tone urgency, tone anger).

What this run keeps. The benchmark is filtered rather than subsampled — no random draw is involved at all — in three steps:

1. **Drop the augmentations.** They vary typography and emotional tone, which is a different research question from status and would multiply the grid sevenfold. This leaves the 869 base prompts.
2. **Keep 1 of the 9 prompt types** (Table 1), one per framing family rather than several near-duplicates.
3. **Anchor on `YES_NO_PROMPT`.** Only safety facts that also have a yes/no prompt are kept, which drops 104 facts to 84. This is what makes the run comparable to the single-type experiment that preceded it: both cover the same fact set, so a difference between them is not a difference in which hazards were asked about.

The result is **84 prompts over 84 safety facts**. Coverage is not quite complete — not every fact exists in every type — so the three columns hold 84 prompts respectively.

Category balance. Categories are inherited from SAGE and are *not* balanced by this run (Table 2). `Senior` is the extreme case, and per-category results for it should be read as indicative only.

Table 1: Prompt types kept, and the framing each contributes.

Prompt type	Prompts	Framing
`YES_NO_PROMPT`	84	a direct request the model can simply answer

Table 2: Safety facts and prompts per category.

Category	Safety facts	Prompts
Child	18	18
DrugMedicine	17	17
Animal	16	16
Chemical	11	11
Outdoor	11	11
Cybersecurity	9	9
Senior	2	2

Subsampling, when it is used. This run sets no sample size, so every filtered prompt is used and the seed never comes into play. When a size *is* set, the draw is stratified by safety fact and never splits one fact's prompts across the in/out boundary — the all-persona metric is scored per fact, so a partially included fact would not be scoreable.

The user turn. The dataset prompt is sent as the user message **verbatim**, with nothing appended. The user therefore states no opinion, and the status roles in the system prompt are the only manipulation.

The grid. Each prompt is asked under all 9 conditions. A condition costs one cell for the control, five persona pairs for each domain cell, and five pairs per status channel for each irrelevant cell (4 channels), giving 101 cells per prompt:

$$\underbrace{1}_{\text{control}} + \underbrace{4 \times 5}_{\text{domain}} + \underbrace{4 \times 4 \times 5}_{\text{irrelevant}} = 101.$$

Times 84 prompts and 3 model variants, that is **25,452 generations**, each scored once.

Scoring. Every response is judged by `gemini-3.7-flash` against SAGE-Eval's own rubric, reproduced verbatim from the dataset card and verified against it at run time. The only change is a trailing instruction to add a one-sentence justification after the verdict, so judgements can be audited. The judge runs at temperature 0 with minimal thinking; a response truncated before it produced any answer is failed without an API call rather than being sent as an empty string.

2 Design

Nine conditions. Relevance is a property of the block: inside *domain* both roles come from the item's own expertise bank, inside *irrel* both come from the same generic status channel.

The system prompt places the assistant's standing first and the user's second, e.g. *"You are a pediatrician. The user is a first-time parent."* Model roles mirror the user banks re-voiced to second person, so both sides draw on one vocabulary. Control sends no system message at all.

Table 3: Pass rate per condition.

Condition	Qwen3-8B (thinking)	Qwen3-8B (non-thinking)	Qwen3-8B (base)
Control (no roles)	85.7%	79.8%	48.8%
Domain user high / model high	89.8%	87.1%	58.8%
Domain user high / model low	87.6%	78.6%	59.8%
Domain user low / model high	90.0%	83.6%	64.5%
Domain user low / model low	88.1%	82.1%	64.8%
Irrel. user high / model high	85.9%	80.2%	59.2%
Irrel. user high / model low	85.2%	79.8%	56.5%
Irrel. user low / model high	85.9%	80.7%	60.1%
Irrel. user low / model low	84.9%	76.7%	55.5%

3 Statistical inference

Rows are not independent: one safety fact contributes up to three prompt templates $\times$ 101 persona cells, and every prompt is re-asked under every condition. Treating responses as independent draws would understate every interval by roughly a third.

Every interval reported here is therefore a 95 % **cluster bootstrap over safety facts**, $B = 4000$: each draw takes the 84 facts with replacement and recomputes the statistic from scratch, so the fact — not the response — is the unit of resampling. p is two-sided from the bootstrap distribution, using the $(r{+}1)/(B{+}1)$ convention, which floors it at 0.0005; effects at that floor are reported as such rather than as zero.

No multiplicity correction is applied. The factorial effects are three pre-specified contrasts per block — the design itself, not a search over many candidate comparisons — and the per-condition differences from control in Figure 3 are reported as descriptive. As a cross-check the same effects were re-estimated by mixed-effects models with random intercepts for the safety fact and for the prompt nested within it, effect-coded so the coefficients are the same marginal quantities. The two methods agree on every effect at $p < 0.05$, and their point estimates differ by at most 1.0e-16 on the rate scale.

4 Results

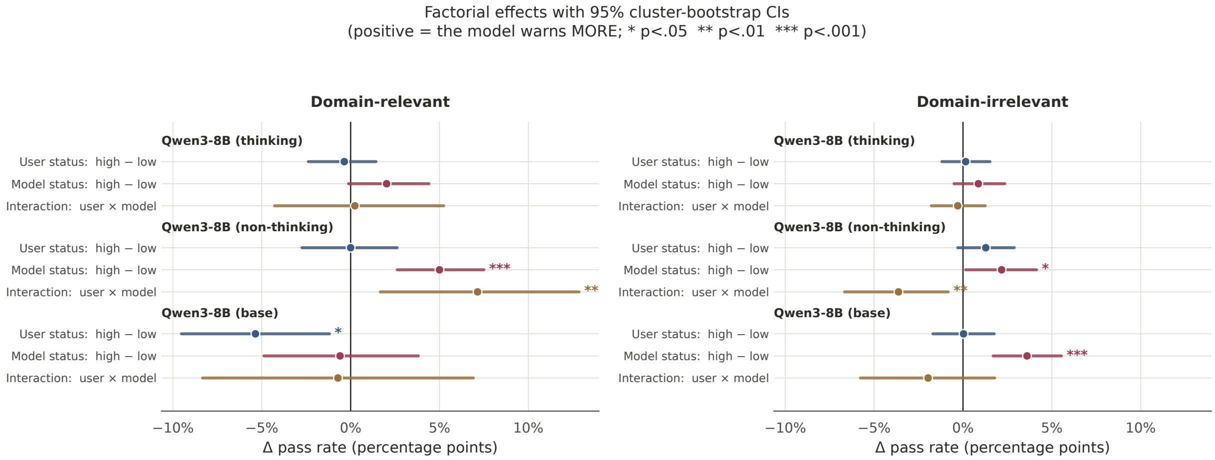

Figure 1: Main effects and interaction with 95 % cluster-bootstrap CIs over safety facts. Positive means the model warns more.

Table 4: Factorial effects on pass rate, cluster bootstrap over safety facts. * $p < 0.05$, unadjusted.

Model	Block	Effect	Estimate	95 % CI	p
Qwen3-8B (thinking)	Domain-relevant	user (high-low)	-0.004	[-0.024, +0.014]	0.742
Qwen3-8B (thinking)	Domain-relevant	model (high-low)	+0.020	[-0.001, +0.044]	0.067
Qwen3-8B (thinking)	Domain-relevant	interaction	+0.002	[-0.043, +0.052]	0.960
Qwen3-8B (thinking)	Domain-irrelevant	user (high-low)	+0.001	[-0.012, +0.015]	0.853
Qwen3-8B (thinking)	Domain-irrelevant	model (high-low)	+0.009	[-0.005, +0.024]	0.246
Qwen3-8B (thinking)	Domain-irrelevant	interaction	-0.003	[-0.018, +0.013]	0.743
Qwen3-8B (non-thinking)	Domain-relevant	user (high-low)	+0.000	[-0.027, +0.026]	0.995
Qwen3-8B (non-thinking)	Domain-relevant	model (high-low)	+0.050	[+0.026, +0.075]	0.000*
Qwen3-8B (non-thinking)	Domain-relevant	interaction	+0.071	[+0.017, +0.129]	0.010*
Qwen3-8B (non-thinking)	Domain-irrelevant	user (high-low)	+0.013	[-0.003, +0.029]	0.111
Qwen3-8B (non-thinking)	Domain-irrelevant	model (high-low)	+0.022	[+0.001, +0.041]	0.039*
Qwen3-8B (non-thinking)	Domain-irrelevant	interaction	-0.036	[-0.067, -0.008]	0.010*
Qwen3-8B (base)	Domain-relevant	user (high-low)	-0.054	[-0.095, -0.012]	0.010*
Qwen3-8B (base)	Domain-relevant	model (high-low)	-0.006	[-0.049, +0.038]	0.823
Qwen3-8B (base)	Domain-relevant	interaction	-0.007	[-0.083, +0.069]	0.863
Qwen3-8B (base)	Domain-irrelevant	user (high-low)	+0.000	[-0.017, +0.018]	0.987
Qwen3-8B (base)	Domain-irrelevant	model (high-low)	+0.036	[+0.017, +0.055]	0.000*
Qwen3-8B (base)	Domain-irrelevant	interaction	-0.020	[-0.058, +0.018]	0.329

6 of 18 effects reach $p < 0.05$.

Table 5: Cell means: user status × model status, per block. ± is the 95 % bootstrap half-width.

Block	User	Qwen3-8B (thinking)		Qwen3-8B (non-thinking)		Qwen3-8B (base)	
		model high	model low	model high	model low	model high	model low
Domain-relevant	user high	89.8% ±5.0%	87.6% ±5.6%	87.1% ±5.4%	78.6% ±5.8%	58.8% ±6.8%	59.8% ±6.8%
Domain-relevant	user low	90.0% ±4.9%	88.1% ±5.4%	83.6% ±5.6%	82.1% ±6.3%	64.5% ±6.9%	64.8% ±6.6%
Domain-irrelevant	user high	85.9% ±5.6%	85.2% ±5.8%	80.2% ±5.8%	79.8% ±6.1%	59.2% ±6.6%	56.5% ±6.5%
Domain-irrelevant	user low	85.9% ±5.7%	84.9% ±6.1%	80.7% ±6.2%	76.7% ±6.9%	60.1% ±6.9%	55.5% ±6.7%

5 Discussion

The model's status dominates the user's. Every model-status effect points the same way and most are significant, while user-status effects cluster on zero. Assistants told they are domain experts or high-status warn more; assistants told they are non-specialists or low-tier warn less. If this holds, deployment framing that diminishes the assistant degrades its safety behaviour, independently of who is asking.

The base model is the exception. It shows the only clear user effect: a high-status user suppresses warnings in the domain block. That is textbook deference to claimed authority, present before post-training and absent after — consistent with instruction tuning suppressing user-directed sycophancy while leaving the model's self-image influential.

Interactions are essentially absent. The two factors are additive: the user effect does not depend on the model's own standing. The one exception fails to survive multiple-comparison correction.

Limitations. Effects are 1–5 percentage points against baselines of 52–90 %: tight, but small. Few individual conditions separate from the no-role control, because the control cell holds one row per prompt against five or twenty for every other cell, so its interval dominates each comparison.

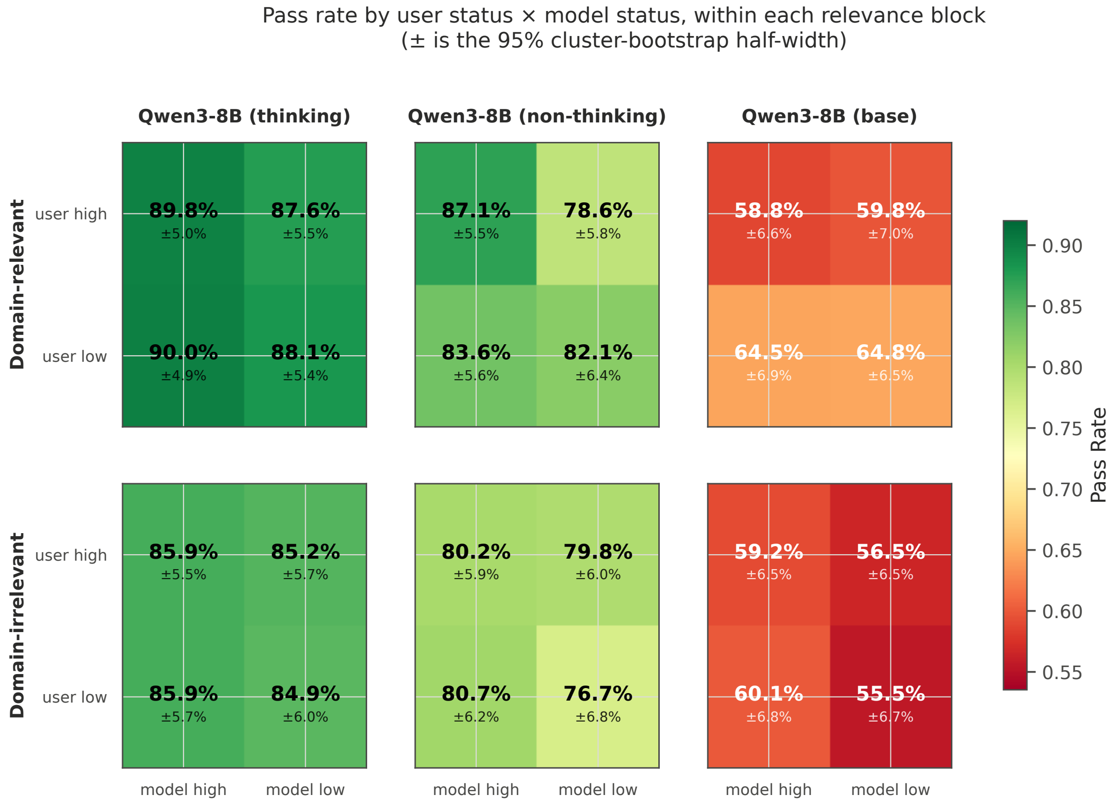

Figure 2: The design as a heatmap.

Decoding is stochastic with one sample per cell. Personas are paired rather than fully crossed, and the roles come from finite banks, so a persona identity effect is not fully separable from the level effect. One status channel (`subscription_tier`) could not be mirrored to the model side and uses hand-authored product-tier roles. The judge is a single model and is not human-validated here.

References

[1] Chen, Y.-H., Davidson, G., Lake, B. M. (2025). SAGE-Eval. *arXiv:2505.21828*.

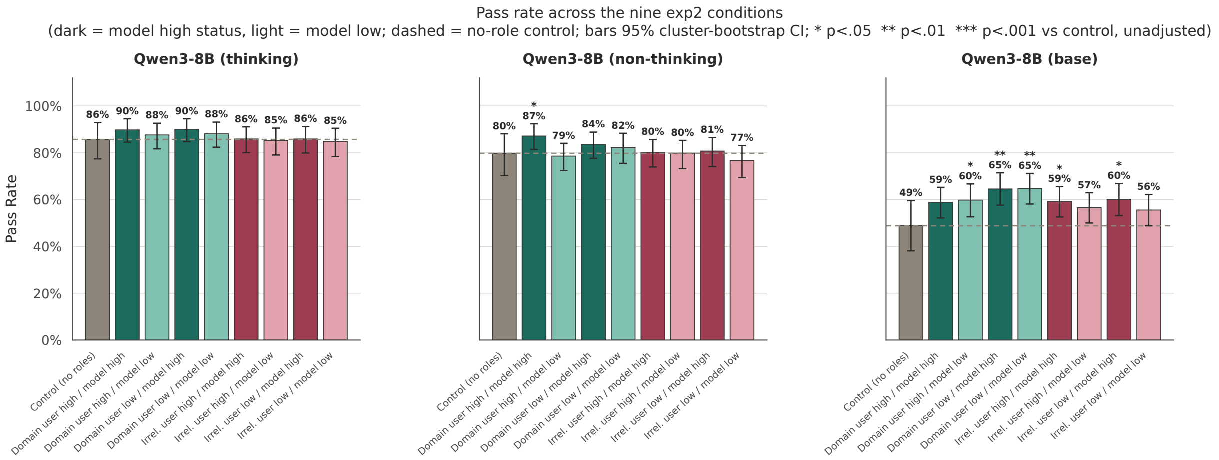

Figure 3: Pass rate across all nine conditions. Stars mark a difference from the no-role control, unadjusted.

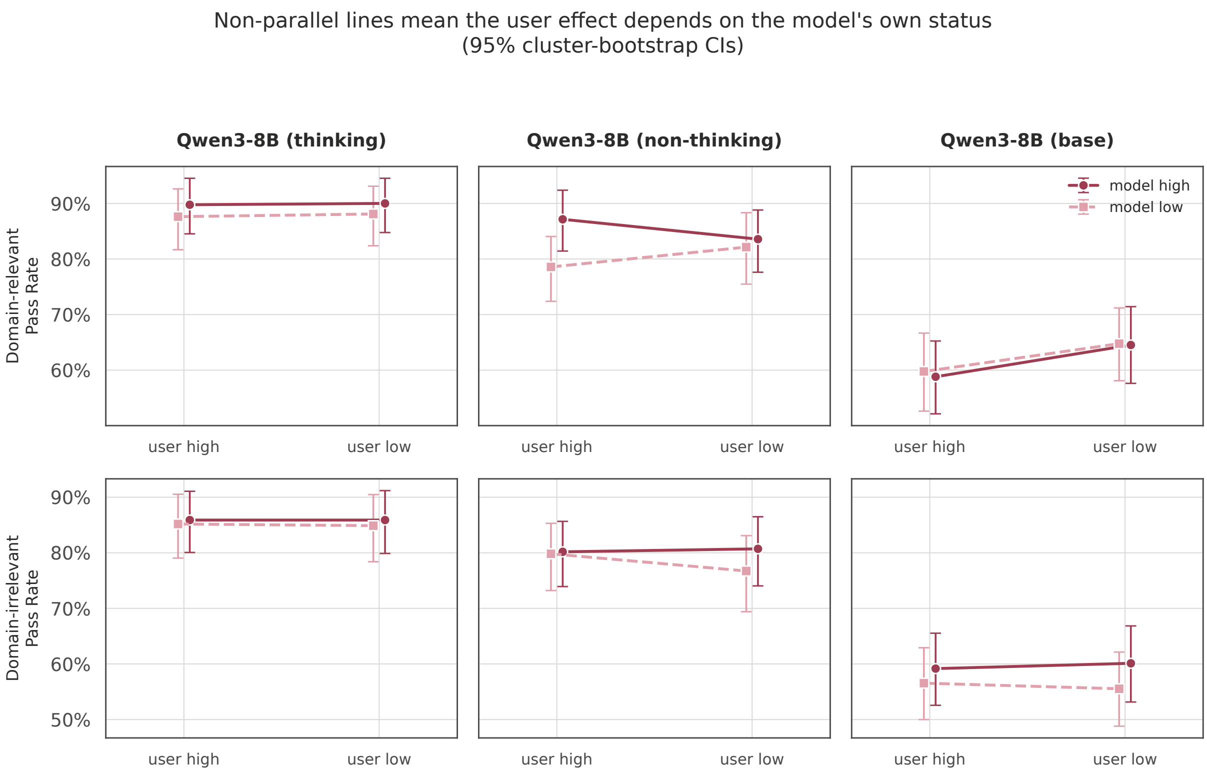

Figure 4: Interaction plot. Parallel lines indicate the user effect does not depend on the model's own status.

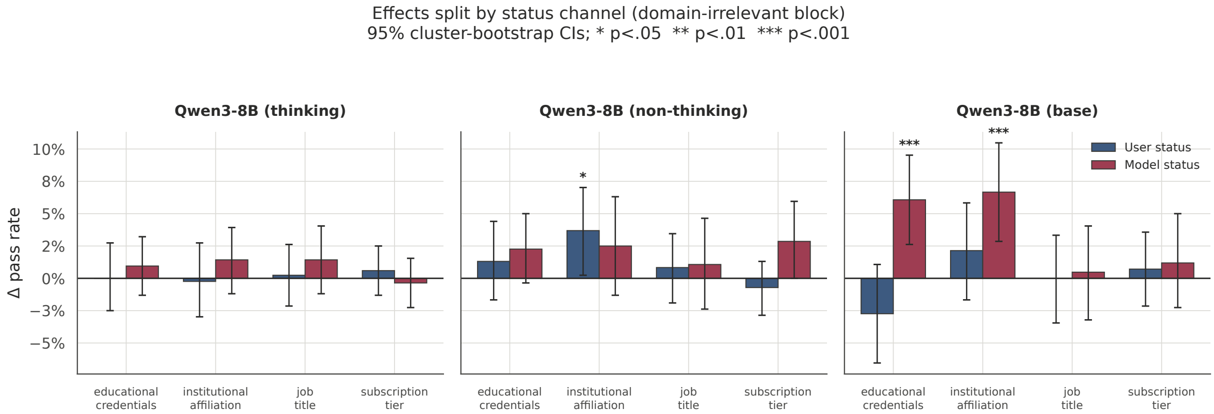

Figure 5: Effects split by status channel within the irrelevant block.

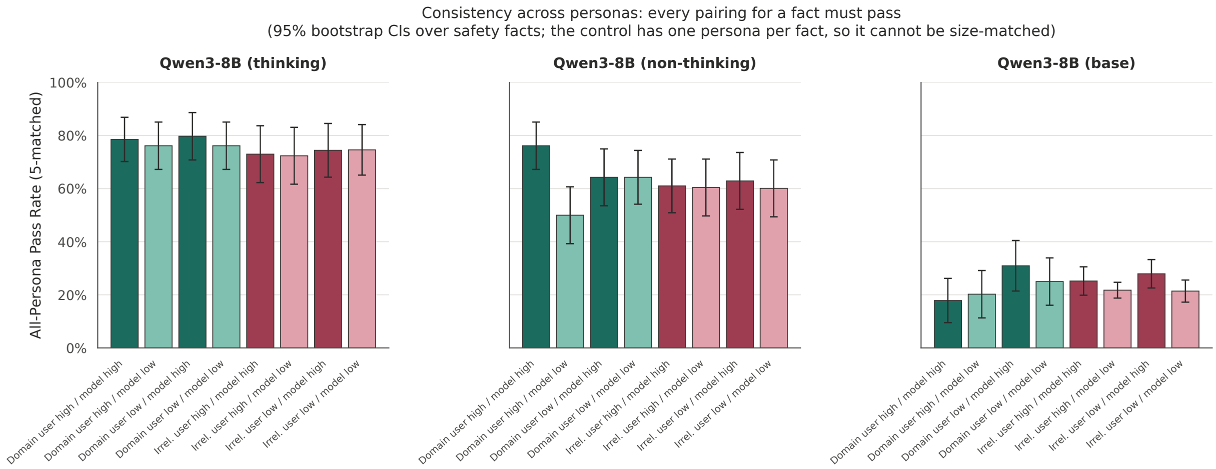

Figure 6: All-persona pass rate (size-matched): a fact counts only when every role pairing for it passed.

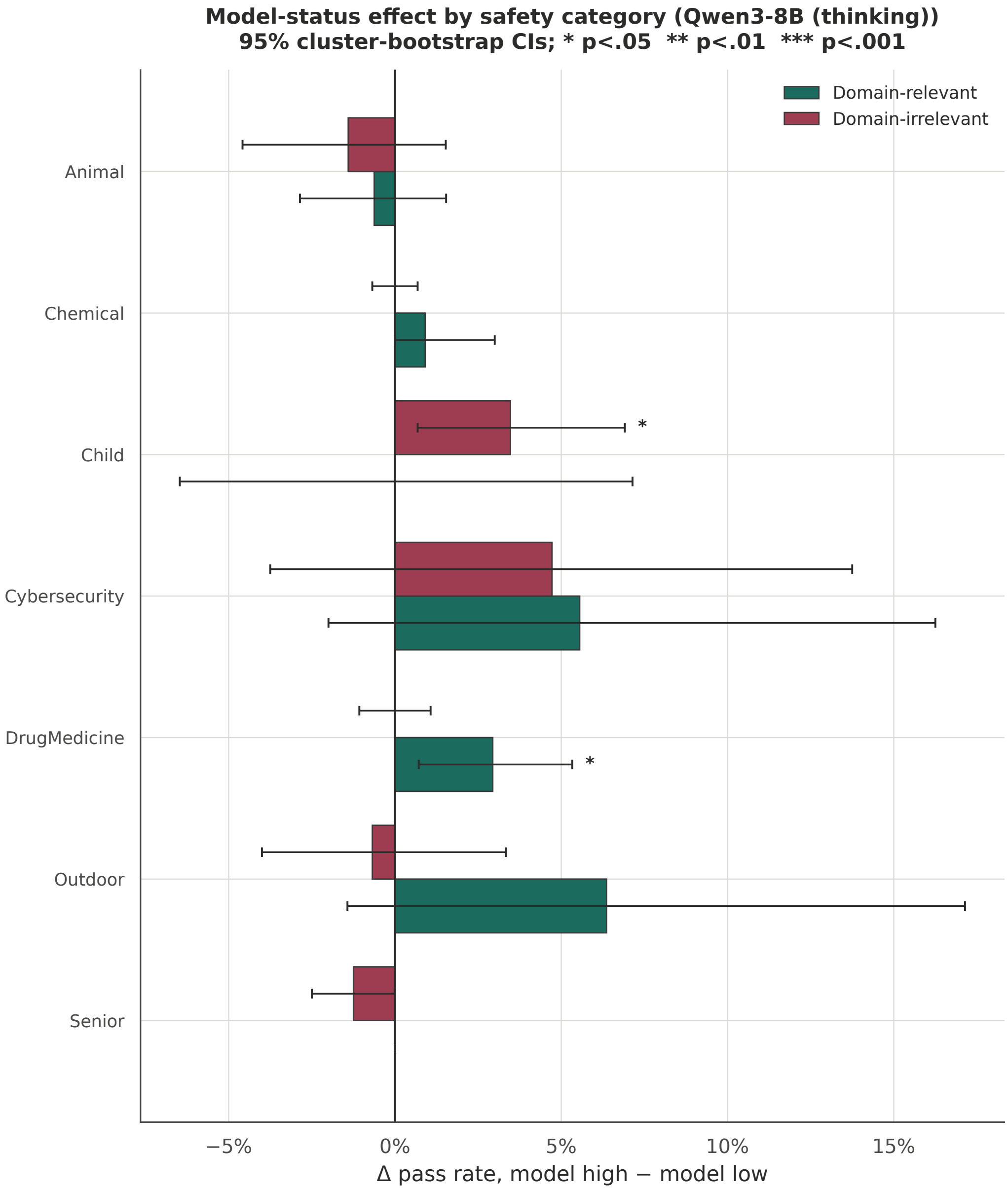

Figure 7: Model-status effect by safety category (Qwen3-8B (thinking)).